

Student Declaration of AI Tool use in this Assessment

Please indicate your level of usage of generative AI for this assessment - please tick the appropriate category(s).

If the “Assisted Work” or “Partnered Work” category is selected, please expand on the usage and in which elements of the assignment the usage refers to.

Solo Work	S1 - Generative AI tools have not been used for this assessment.	☐
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.	☐
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student’s own work.	☐
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student’s own work.	☐
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student’s responsibility.	☐
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.	☐
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.	☐
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.	☐
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.	☐

	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.	☐
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.	☐
	A11 - Other uses not listed above Please specify:	☐
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify: AI was used to answer queries I had about errors and non-functioning pieces of code.	☒

Please provide details of AI usage and which elements of the coursework this relates to:

Why is it that changing my include from glew to glad causes errors in glew, when it is not being used?

GitHub Copilot

▶ 1 reference

Short answer: because the project is still building and/or linking GLEW somewhere else. Even if this file stops including GLEW, other translation units, your precompiled header, or your project/linker settings can still pull GLEW in. Also, your code still calls **glewInit()**, which requires GLEW.

What to fix

1. Use only one OpenGL loader. If switching to GLAD:

Remove all **#include <GL/glew.h>** across the project (search solution-wide).

Remove calls to **glewInit()**; initialize GLAD instead.

Ensure include order: include GLAD before GLFW.

Add GLAD source to the project and link settings appropriately.

Why is my shark using the rock texture instead of its own?

GitHub Copilot

▶ 3 references

▶ 2 tasks done

Short answer: because nothing rebinds a new diffuse texture for the shark, your fragment shader keeps sampling the diffuse texture that Rock bound on texture unit 0.

What's happening

Your model shader always samples **texture_diffuse1**.

Rock.Draw(ModelShaders) binds its diffuse texture to unit 0 and sets the sampler to 0.

If Assimp didn't find a diffuse texture for the shark (missing/incorrect MTL/paths), **Shark.Draw(ModelShaders)** won't bind anything new. The previously bound rock texture on unit 0 remains active, so the shark samples the rock texture.

How to verify quickly

After drawing the rock, unbind the texture and draw the shark. If the shark turns black/solid, it confirms the shark had no diffuse bound and was reusing the rock's.

I want Shark to move instead of the camera when I press wasd. How do I go about changing this?

GitHub Copilot

▶ 1 reference

Take a breath and make the Shark an object you move, not the camera. Add a world-space position for the Shark, update that from WASD, and use it when building the Shark's model matrix. Also reset **model** per draw to avoid compounding transforms.

1. Add a Shark position

main.cpp

Apply

//Relative position within world space
vec3 cameraPosition = vec3(3.0f, 2.0f, 3.0f);
//The direction of travel
vec3 cameraFront = vec3(0.0f, 0.0f, -1.0f);
//Up position within world space
vec3 cameraUp = vec3(0.0f, 1.0f, 0.0f);

// Add this:
vec3 sharkPosition = vec3(0.0f, 0.0f, 0.0f);

2. Move the Shark in **ProcessUserInput** instead of the camera

My square does not seem to have the image "absolutemichael.jpg" on it. Why on earth is this happening?

GitHub Copilot

▶ 3 references

▶ 2 tasks done

Short answer: you're binding a texture, but you never use it in the shader, and you're not actually drawing a VAO that contains the quad's vertices/UVs.

What's going on

The active shaders don't sample a texture. **shaders/fragmentShader.frag** just outputs **colourFrag**; there's no **sampler2D** or UVs in either shader.

The VBO/VAO you draw belongs to the terrain (position + color only). The quad data (with UVs) is never uploaded to a buffer or referenced by a VAO.

You call **glDrawElements(GL_TRIANGLES, 6, ...)** with the terrain VAO still bound; that just draws the first two terrain triangles, not your quad.

You set a uniform on **program** (from **LoadShaders**) but render with **Shaders.use()**. That uniform write may target a different program object.

Minor: you don't set the sampler uniform to a texture unit, and you don't enable a texcoord vertex attribute.

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒